

→

$$T = 6.2 \text{ s}$$

9

C

The gravitational field strength at P due to both masses are equal and in opposite directions → resultant field strength = 0

$$\phi = \phi_m + \phi_m = -\frac{Gm}{r/2} - \left(-\frac{Gm}{r/2} \right) = -\frac{4Gm}{r}$$

10

D

Gravitational force provides the centripetal force for the satellite to orbit the Earth